



**Verified Carbon
Standard**

IMPROVED COOKSTOVE INITIATIVE IN KARATU DISTRICT, TANZANIA

T A N J A

Project title	<i>Improved Cookstove Initiative in Karatu District, Tanzania</i>
Project ID	5692
Crediting period	<i>01 - July - 2025 to 30 - June - 2032</i>
Original date of issue	<i>To be filled after completion of the audit</i>
Most recent date of issue	<i>18-August - 2025</i>
Version	<i>1.0</i>
VCS Standard Version	<i>Version 4.7</i>
Prepared by	<i>TANJA Corporation Limited</i>

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	4
1.3	Sectoral Scope and Project Type	5
1.4	Project Eligibility	5
1.5	Project Design	7
1.6	Project Proponent	8
1.7	Other Entities Involved in the Project	9
1.8	Ownership.....	9
1.9	Project Start Date	9
1.10	Project Crediting Period	9
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	10
1.12	Description of the Project Activity	10
1.13	Project Location	12
1.14	Conditions Prior to Project Initiation	13
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	13
1.16	Double Counting and Participation under Other GHG Programs	14
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	14
1.18	Sustainable Development Contributions	15
1.19	Additional Information Relevant to the Project	16
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	16
2.1	Stakeholder Engagement and Consultation.....	16
2.2	Risks to Stakeholders and the Environment.....	24
2.3	Respect for Human Rights and Equity	26
2.4	Ecosystem Health	29
3	APPLICATION OF METHODOLOGY.....	31
3.1	Title and Reference of Methodology	31
3.2	Applicability of Methodology	32
3.3	Project Boundary	35
3.4	Baseline Scenario	36

3.5	Additionality	36
3.6	Methodology Deviations	38
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	38
4.1	Baseline Emissions	38
4.2	Project Emissions	38
4.3	Leakage Emissions	38
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	38
5	MONITORING	40
5.1	Data and Parameters Available at Validation	40
5.2	Data and Parameters Monitored.....	42
5.3	Monitoring Plan.....	46

1 PROJECT DETAILS

1.1 Summary Description of the Project

This initiative focuses on the distribution of high-efficiency improved cooking stoves (ICS) across Tanzania. The program aims to replace existing, inefficient stoves with new models that boast superior thermal efficiency. These stoves utilize wood more effectively, ensuring better heat transfer to cookware, thus conserving fuel and reducing emissions of greenhouse gases. By curbing the need for excessive firewood, the project also addresses issues like deforestation, indoor air pollution, and the considerable time people spend gathering wood.

Given that the targeted beneficiaries may not have the means to afford these stoves, the initiative typically involves significant subsidies or outright donations. Project proponents are made aware that the operation of these stoves generates carbon credits, which helps finance the stoves' costs and cover project operational expenses.

Specific details for the initial activities are outlined in Section 1.12 of the document, with information and KML files for expansions into other countries to be supplied in the document upon each activity instance's verification stage.

The introduction of ICS is set to replace traditional open-fire cooking methods, enhancing wood fuel efficiency and thereby saving on fuel and cutting back on greenhouse gas emissions. Without the project, the target population would continue relying on non-renewable wood for their thermal energy needs, a baseline scenario this project aims to improve upon.

For the initial phase, the project plans to distribute 5,000 ICSs; however, the number of units distributed in the future may change depending on the status of implementation and will be updated as appropriate during the registration and verification phase of project activities. The project plans to distribute 5,000 units for seven years. The projected GHG reduction from the distribution is estimated at 11,008 tCO₂e for 5,000 cookstoves.

1.2 Audit History

For projects undergoing crediting period renewal, include the audit history of the project using the table below. For the project validation, state the validation date in the Period column. This table should include all monitoring periods, including the period of this report.

Audit type	Period	Program	Validation/verification body name	Number of years
------------	--------	---------	-----------------------------------	-----------------

Validation	We plan to conduct the audit in 2025.	VCS	Earthood Services Limited	One year
Verification	We will provide an update once the verification schedule is finalized.	VCS		

1.3 Sectoral Scope and Project Type

Complete the table below with information relevant for non-AFOLU projects:

Sectoral scope¹	03 Energy demand
Project activity type	Type:II Energy efficiency improvement projects

Complete the table below with information relevant for AFOLU projects:

Sectoral scope	Not applicable
AFOLU project category²	Not applicable
Project activity type	Not applicable

1.4 Project Eligibility

1.4.1 General eligibility

The project fits within the VCS Program's category for thermal-application efficiency upgrades and is not among the activities excluded in Table 1 of the VCS Standard v4.7. Moreover, the methodology applied—VM0050 v1.0—qualifies because it was both developed and approved by Verra. Please note that some activities may still be ruled out if they meet the specific exclusion conditions detailed in Table 1 of the VCS Standard v4.7.

Activity	Exclusion	Applicability
Grid-connected electricity generation activities using hydroelectric power plants.	Excluded in non-LDCs. Further, large-scale projects excluded in LDCs ⁴	Not applicable. The project, including all activity instances, does not involve the use of hydroelectric power facilities.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

Grid-connected electricity generation activities using wind, geothermal or solar photovoltaic (PV) power plants.	Excluded in non-LDCs	Not applicable. The project centers on distributing improved cookstoves to enhance thermal-energy efficiency.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or Trigeneration	Excluded in non-LDCs	Not applicable. Neither the project nor any of its activities employs waste-heat recovery techniques.
Activities generation electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass or forest residue biomass	Excluded in non-LDCs	Not applicable. The project does not generate electricity or industrial heat from non-renewable biomass, agro-residues, or forest residues; its sole aim is to supply improved cookstoves for household thermal needs.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	Excluded in non-LDCs	Not applicable. No fossil fuels are used, and the project does not entail any fuel-switching measures.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	Excluded for large-scale projects in non-LDCs	Not applicable. The project and its activities rely on improved cookstoves and do not substitute any electric lighting systems.

Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	Excluded for large-scale projects in non-LDCs	Not applicable. The project does not involve the installation or replacement of electricity transmission infrastructure.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Excluded in all countries	Not applicable. The project and its activity instances do not target reductions of HFC-23 emissions.

1.4.2 AFOLU project eligibility

Not applicable. This project is not an AFOLU project.

1.4.3 Transfer project eligibility

Not applicable. This is a new project and is not transferred from another GHG program.

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

Project Description for grouped projects:

S.No.	Criteria for Grouped Projects	Justification for Compliance with Grouped Project Activity
-------	-------------------------------	--

1	Clearly defining the geographic area(s) where all project activity instances will take place, using geodetic polygons as specified in Section 3.11.	Project activities take place inside clearly defined Tanzania's boundaries, charted with geodetic polygons in line with VCS Standard 3.11. These polygons cover communities that still rely on traditional cooking practices, enabling precise monitoring and preventing overlap with any other programs.
2	Establishing one or more baseline determinations for the project activity in line with the methodology requirements.	The baseline assumes households would keep using inefficient, traditional cookstoves fueled by non-renewable biomass. Household surveys across the project zone confirmed this widespread practice before rollout, and the baseline was set in accordance with Section 6 of methodology VM0050.
3	Demonstrating additionality of the project activity in accordance with the methodology's requirements.	Additionality is demonstrated via the activity-method in the applied methodology: no government mandate requires improved stoves, participation is voluntary, and the project's feasibility depends on revenues from GHG credits—thereby showing regulatory surplus and meeting the positive-list criteria.
4	Defining eligibility criteria for adding new project activity instances at future verification events.	Eligibility rules have been established to align with the methodology and project objectives. These cover geographic limits, approved technologies, and compliance with baseline and additionality requirements. A dedicated database logs every new activity instance to prevent duplication or double counting.
5	Describing the central GHG information system and controls for project monitoring.	A central GHG information system stores all project data. Each stove is given a unique serial number captured through a TANJA's project database sheet that records its distribution, installation, and use. This sheet boosts transparency, supports robust monitoring and verification, and integrates field visits and user feedback to keep data accurate.

1.6 Project Proponent

Organization name	TANJA Corporation Limited
Contact person	Tomoko Ogo
Title	Senior Project Manager
Address	OLDEANI Ward, Karatu District, Arusha Region Tanzania
Telephone	+255 760 957 366
Email	cookstove.project@tanja.co.tz

1.7 Other Entities Involved in the Project

1.8 Ownership

The project ownership is with TANJA Corporation.

The ICS are distributed to end uses (Households) at a price of TZS 10,000 per unit. This contribution is below cost and generates no profit for TANJA. The end users are informed in advance that the use of ICS generates carbon finance which in turn is used to cover the price of ICS and for recovering project implementation costs. The participating household will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to TANJA Corporation, Also undertaking is provided by TANJA Corporation for projects ownership.

1.9 Project Start Date

Project start date	01-July-2025
Justification	This is the date when the distribution of cookstoves began to generate GHG emission reductions.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	01-July-2025 to 30-June-2032

First crediting period: 01- July-2025 to 30- June-2032, seven years renewable crediting period. Being renewable crediting period, total crediting period of grouped project activity is 14 years.

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-July-2025 to 30-June-2026	127
01-July-2026 to 30-June-2027	667
01-July-2027 to 30-June-2028	832
01-July-2028 to 30-June-2029	1,546
01-July-2029 to 30-June-2030	2,747
01-July-2030 to 30-June-2031	2,610
01-July-2031 to 30-June-2032	2,479
Total estimated ERRs during the first or fixed crediting period	11,008
Total number of years	7
Average annual ERRs	1,572

1.12 Description of the Project Activity

This initiative is focused on deploying fuel-efficient improved cooking stoves (ICS) to households, aimed at replacing conventional traditional stoves. The ICS selected for this program are designed for high energy efficiency, significantly lowering both fuel usage and emissions during cooking and water heating activities in residences. By introducing advanced energy-efficient technologies, these stoves enhance combustion efficiency and heat transfer to cooking pots, outperforming traditional three-stone fires or basic pot supports. This leads to a marked decrease in the amount of wood fuel needed, compared to traditional methods or less efficient stove designs.

Detailed specifications for these stoves will be made available at the time of validation/verification or when specific instances of project activities are incorporated.

For the initial instance of project activity, technical details for the cook stoves from BURN will be provided:

Manufacturer BURN:

DETAILS			
A)	Stove Manufacturer	BURN	
	Stove Model	Ecoa Wood Generation 3	
	Stove type	Firewood	
B)	Stove Material Used	Stove Body	Stainless Steel
		Pot Rest	Stainless Steel
		Burning Chamber	Stainless Steel
		Ash Tray	Stainless Steel
		Feet	Aluzinc
		Handles	Stainless Steel
C)	Measurements	Height	28.5 cm
		Diameter (stove top)	26.9 cm
		Weight	3.4 kg
		Fuel Chamber Volume	1,288 cm ³
		Packaging Dimension	29L x 29W x 29H cm
D)	WBT Results	Thermal Efficiency	50.8 %
		High power Output	2.8 kW
		High power PM2.5	268.72 Mg/MJd
		High power CO	4.42 g/MJd
		High power indoor emissions CO	0.50 g/min
		High power indoor emissions PM2.5	24.56 Mg/min
		Turndown ratio	1.55
E)	Expected life of stove	7-10 Years	

The manufacturer for future project activity instances may change, and any changes will be transparently reflected during verification. The details and specifications of the respective cookstoves will be considered for the calculation of estimated GHG emission reductions or removals for each project activity instance.

Type of ICS used is based on below thermal efficiency:

Tier 5 Thermal Efficiency $\geq$ 50%

Tier 5 refers to the ISO Voluntary Performance Targets (ISO/TR 19867-3) for clean cookstoves.

Data collection of ICS end-user

The project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process along with the following information (as appropriate and available):

- 1) Name of ICS user or head of the household
- 2) Address/ Village name of ICS household
- 3) Stove model Distributed
- 4) Date of distribution/installation
- 5) ICS serial number
- 6) Retailer/distributor/Manufacturer information

The information collected will be stored in an electronic database, specifically an Excel sheet, which will serve as the project database for monitoring and sampling purposes.

1.13 Project Location

All project activity instances in the proposed grouped project will be located within the geographical boundaries of Tanzania. For the project activity instances, the geographical area includes six wards in Karatu (Oldeani, Daa, Qurus, Ganako, Endabash and Endamarariiek), though it will eventually expand to other areas in Karatu district. The KML file is provided below. The geographical boundary for projects located in Tanzania is delineated using geographic coordinates of the project areas as follows:

Latitude – -3.386925°

Longitude – -36.682995°

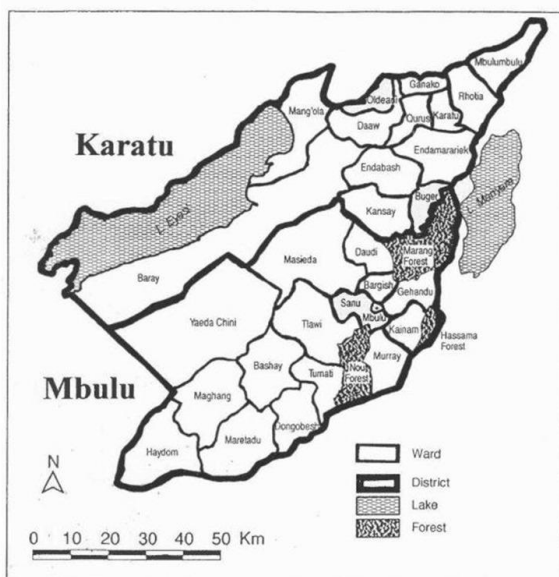


Figure: Location map of the study area

1.14 Conditions Prior to Project Initiation

The objective of the project is to reduce the reliance on fossil fuels and biomass for domestic cooking. According to the World Bank Group, “Clean Cooking Background Note – Background Paper for the United Republic of Tanzania Country Climate and Development Report” (November 2024), more than 89 % of households on mainland Tanzania still depend on traditional polluting fuels and technologies for cooking. Within rural areas—the focus of the TANJA Project—this reliance remains exceptionally high, with 84.8 % of households chiefly using firewood, compared with only 17.4 % in urban centres. Therefore, the 84.8 % rural figure is the most appropriate value for the project’s baseline scenario. By introducing energy-efficient stoves that utilize wood and charcoal, this project aims to mitigate the environmental and societal impacts associated with the inefficient use of biomass for cooking, with a focus on low-income families. The conditions prior to the start of this project, involving the continued use of three-stone stoves serve as the baseline scenario.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

In Tanzania, there are no specific legal or regulatory frameworks in place that promote the adoption of improved cooking stoves in household settings. The initiative to deploy these stoves is a voluntary effort taken by the project proponent. While the Tanzanian government promotes for the use of cleaner fuels, such as LPG, the high initial and ongoing costs associated with it have resulted in low adoption rates, leaving the majority of households to continue relying on traditional cooking methods. Applicable general laws exist; however, no project-specific legal requirements for improved cookstove distribution or related carbon activities have been identified. The project will comply with all applicable general requirements and will monitor any legal or regulatory changes.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

The project offers a myriad of benefits spanning social, environmental, economic, and technological aspects, contributing to the sustainable development of both the local community and the nation at large. Here are the benefits detailed in another way:

Social Benefits

Reduce the burden on women and other family members by cutting firewood use and the time spent collecting it.

Improve indoor air quality by decreasing smoky cooking.

Lower the risk of respiratory illness associated with indoor smoke.

Enable quicker meal preparation.

Keep kitchens cleaner with far less soot, improving overall family well-being.

Environmental Benefits

Reduce reliance on firewood (non-renewable biomass).

Help protect local forests.

Lower greenhouse gas emissions.

Decrease the water and effort needed to clean cookware due to a cleaner cooking process.

Economic Benefits

Create local job opportunities through project implementation (e.g., stove distribution and monitoring).

Save households money on cooking fuel.

Free up time otherwise spent collecting firewood, allowing family members to engage in income-generating or other productive activities.

The project also contributes to SDG 3 (Health), SDG 5 (Gender Equality), SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action). The indicators, methods and frequencies summarized are aligned with the project's monitoring plan in Section 5.3 (Monitoring Plan).

SDG 3: Good Health and Well-being

· Contribution: Reduced indoor air pollution leading to lower risk of respiratory diseases

SDG 5: Gender Equality

- Contribution: Reduced time spent collecting firewood, especially for women and children, increasing opportunities for education and economic activities

SDG 7: Affordable and Clean Energy

- Contribution: Improved energy efficiency and expansion of sustainable cooking methods

SDG 13: Climate Action

- Contribution: Mitigation of climate change via reduced GHG emissions from ICS adoption

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The stakeholders identified for this project include:

Primary Target Areas

- Six wards in Karatu (including Oldeani, Daa, Qurus, Ganako, Endabash and Endamarariek).

Local Communities

	·Residents in these six wards are the primary stakeholders and intended beneficiaries of the improved cookstoves (ICS) distribution. Representatives from women’s groups and community leaders, who are also residents. Employees and Workers ·Individuals employed by TANJA’s agricultural operations, many of whom are from local communities. Local Government Representatives ·Authorities involved in approving, facilitating, and overseeing the project activities. Key participants in last year’s SHM, which primarily included government officials and community representatives. NGOs ·NGOs focusing on environmental conservation, community development, or both. Their potential roles include project monitoring, capacity building, and facilitating stakeholder consultations. Women’s Group Representatives and Community Leaders ·Invited to last year’s SHM and actively involved in discussions. They serve as critical links to broader community engagement, particularly for women’s empowerment and local advocacy. Note: Specific individuals or organizations can be listed after further stakeholder consultations.
Legal or customary tenure/access rights	·Land Ownership: Although TANJA holds property rights to part of the Oldeani ward, one of the project areas, the project does not involve land acquisition, land-use change, or any restriction on community access to natural resources. Activities consist of the distribution and household use of improved cookstoves and related monitoring. ·TANJA's agricultural operations adhere to local and national regulations governing land use, ensuring no conflict with indigenous or community rights.

	•The project does not restrict any stakeholder from accessing resources and is designed to complement existing legal and customary frameworks.
Stakeholder diversity and changes over time	The stakeholders exhibit diversity in terms of: •Socioeconomic factors: Some community members are employed in TANJA's agricultural business, while others rely on subsistence farming and local labor markets. •Anticipated changes: Over time, the adoption of ICS is expected to improve health outcomes (reduction in indoor air pollution), economic savings (reduced firewood expenditure), and social cohesion through shared community benefits. Example assumption: A future assessment will track how different groups adapt to the introduction of ICS to ensure equitable benefits. •Social and Gender Empowerment When women, children, and families spend less time gathering firewood, they can reallocate that saved time to social or economic pursuits, such as education, small-scale enterprises, or community development activities. The burden of collecting firewood is particularly heavy during the rainy season, as damp wood is harder to ignite and requires larger quantities. ICS usage alleviates this challenge, freeing women and children from additional labor and reducing their exposure to harmful smoke. TANJA will review and update the stakeholder diversity and changes at each verification.
Expected changes in well-being	•Economic Benefits Reduced household expenditures on firewood, allowing families to allocate more funds toward other needs. Less time spent on wood-gathering translates to increased opportunities for income-generating or other productive activities. •Health Improvements Substantial reduction in indoor air pollution, leading to fewer respiratory illnesses. Because ICSs require less fuel and have high

	combustion efficiency, users are exposed to significantly less smoke—especially critical during the rainy season when it is difficult to gather wood, or when cooking must be done only indoors. ·Environmental Benefits By using less firewood overall, ICSs contribute to more sustainable resource management and protection of local forests. ·Social and Gender Empowerment When women, children, and families spend less time gathering firewood, they can reallocate that saved time to social or economic pursuits, such as education, small-scale enterprises, or community development activities. The burden of collecting firewood is particularly heavy during the rainy season, as damp wood is harder to ignite and requires larger quantities. Because ICS burn more efficiently than three-stone stoves, households need less fuel and fewer collection trips. This is expected to reduce the additional labor borne by women and children and to lower exposure to harmful smoke.
Location of stakeholders	The primary stakeholders are located in the six wards of Karatu District, including Oldeani, Daa, Qurus, Ganako, Endabash and Endamarariiek. These areas are rural communities where households predominantly rely on traditional three-stone cooking methods.
Location of resources	The project supplies energy-efficient Improved Cookstoves (ICS) to households in six wards of Karatu District—Oldeani, Daa, Qurus, Ganako, Endabash, and Endamarariiek. It does not utilize or affect any land or resources owned by, or customarily available to, stakeholders; all such property and customary rights remain entirely with their respective owners.

2.1.2 Stakeholder Consultation and Ongoing Communication

Use the table below to describe the process for and the outcomes from the stakeholder consultation conducted prior to project initiation.

Date of stakeholder consultation	26-July-2024
----------------------------------	--------------

Stakeholder engagement process

To ensure culturally appropriate engagement, TANJA conducted a community consultation meeting on 26-July-2024 in the six target wards of Karatu District (Oldeani, Daa, Qurus, Ganako, Endabash and Endamarariek).

·Announcement:

1)Invitation letters and leader outreach.

Swahili invitation letters were hand-delivered to ward/village leaders (WEOs/VEOs and chairpersons) between 24–26 June 2024, with follow-up visits and phone calls by TANJA staff. Leaders also shared the notice via WhatsApp groups to broaden local reach.

2)Attend community meetings

The announcement was reiterated at community/ward meetings convened by local authorities to brief residents ahead of the SHM

3)Newspapers

On 26 June 2024, TANJA placed public announcements in MWANANCHI and UHURU inviting all interested community members to the consultation. The contents (headline “TANGAZO KWA UMMA”) summarized the ICS project and listed the venue/date/time: SUMAWE Hall, Sumawe Complex, Mzizima Street, Karatu; 26 July 2024, 09:30–15:00. Contact details were provided.

4)Email and online outreach

International and local NGOs were invited by email on 25–26 June 2024. The email included the Project Summary and a Feedback Form and invited written feedback during a 30-day window ending 26 July 2024 for stakeholders unable to attend in person.

·Process: The consultation included:

1.Presentations on the project goals, including environmental benefits and health improvements.

2.Group discussion and open Q&A session for addressing concerns and gathering input.

3.A demonstration of the improved cookstoves (ICS). (Cooking makande and tasting)

	·Documentation: Minutes of the meeting were recorded, and attendance was logged with signatures or marks for those unable to sign. Photos and videos were taken as additional records. For stakeholders unable to attend in person, the project provided off-site access and feedback channels: -project information on TANJA's website (non-technical summary); -a 30-day feedback window ending 26 July 2024; -submissions accepted via email, WhatsApp/call, and an online Feedback Form. All submissions were logged and archived together with the minutes.
Consultation outcome	The consultation generated strong community support for the project, with the following key outcomes: ·Consent: Stakeholders expressed broad support for the project, acknowledging the benefits of improved health and reduced fuelwood consumption. During the SHM, it was explained that the benefit-sharing scheme would be reviewed and developed in the future. After the SHM, each Ward Executive Officer (WEO), Village Executive Officer (VEO) (among others) was visited, and the explanation was provided again. ·Risks and Costs: Concerns about stove affordability were addressed by explaining the subsidized distribution model and carbon credit funding. ·Benefits: Participants appreciated the time savings for firewood collection and reduced indoor air pollution, particularly benefiting women and children. ·Laws and Regulations: The project complies with applicable local regulations and relevant international standards, including worker protections under Tanzanian law. TANJA commits to continued compliance and maintains records evidencing such compliance. ·Validation and Verification: Stakeholders were informed about the validation and monitoring processes, ensuring transparency and accountability.
Ongoing communication	To maintain regular communication with stakeholders, TANJA has established the following mechanisms:

	·Community representatives: WEOs and VEOs were appointed as liaisons in each ward to act as points of contact; where WEOs/VEOs were unavailable, other designated local officers served in this role. ·Periodic updates: Annual meetings to provide progress updates and gather feedback. ·Complaint resolution: A dedicated phone line and office hours at TANJA office for addressing any issues related to the project or TANJA's website.
Stakeholder input	All input received during the consultation was considered: • Feedback on stove design: Some participants suggested modifications for larger family sizes. These suggestions will be incorporated into future procurement if feasible. • Requests for broader distribution: While the current project focuses on six wards, plans for future expansion to other areas in Karatu were explained. No major design changes were deemed necessary at this stage as the project already aligned well with community needs.

2.1.3 Free Prior and Informed Consent

Use the table below to describe the outcome of the FPIC process as part of the stakeholder consultation process.

Obtaining consent	Obtaining Consent TANJA obtained Free, Prior, and Informed Consent (FPIC) from the community members in the six target wards (Oldeani, Daa, and others) through a transparent and inclusive consultation process conducted on 26-July-2024. 1. Process for Consent Acquisition: ○ Engagement: Meetings were held in each ward, involving local leaders, community representatives, and other key stakeholders. These meetings were conducted in Swahili to ensure accessibility for all community members.
-------------------	---

	○ Transparency: The objectives, benefits, and potential risks of the project were clearly explained, including how the cookstove distribution would be financed through carbon credits. ○ No objections were raised: Several comments expressed a desire for the project to proceed well, and the SHM was conducted in an overall atmosphere that moving forward would be acceptable. All participants signed the attendance sheet, and the records are kept at the TANJA office. 2. Conflict Resolution: ○ No unresolved conflicts were identified during the consultation process. All community concerns, such as stove affordability and prioritization of recipients, were addressed during the meetings. ○ TANJA has committed to resolving any future concerns promptly through the established complaint resolution mechanisms.
Outcome of FPIC	1. Transparent Agreement: ○ Community members expressed their full support for the project, agreeing to participate in the cookstove distribution program. The community appreciated the emphasis on improving health outcomes and reducing deforestation. ○ A mutual understanding was reached regarding the allocation of cookstoves, prioritizing households with the greatest need, such as those relying on traditional three-stone stoves. 2. Information Disclosure: ○ Detailed information about the project, including the environmental and social benefits, was disclosed. Community members were informed about their rights to accept or decline participation without any coercion. 3. Assurance of Non-Displacement:

	○ The project does not involve any land acquisition or physical relocation. TANJA confirmed that the project activities align with existing land use practices and customary rights. No physical or economic displacement of community members is involved.
--	---

2.1.4 Grievance Redress Procedure

Development process	TANJA keeps a stakeholder register at the TANJA office in Karatu to record feedback and grievances. A contact number and office hours are available for stakeholders to raise complaints or concerns. The TANJA team promptly receives, logs, and addresses grievances, applying culturally appropriate approaches to resolve issues fairly and transparently and all records are retained for audit.
Grievance redress procedure	Developed in consultation with stakeholders to ensure transparency and accessibility. A grievance register and contact number are maintained at the project office. Grievances may be submitted in person, by phone, or in writing (email/letter). Every submission is acknowledged and addressed within a reasonable timeframe, and unresolved matters are escalated to TANJA management. Periodic summaries of grievances and resolutions are shared at stakeholder meetings and via the TANJA office/website to ensure transparency.

2.1.5 Public Comments

Comments received	Actions taken
Will be confirmed once public commenting period is complete.	

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

TANJA has significant experience in managing large-scale operations and engaging local communities through its agricultural activities, which include coffee in the Karatu District. These operations have created employment opportunities, fostered sustainable economic growth, and strengthened relationships with the local community. This foundation positions TANJA well to implement the ICS distribution project, which aligns with its mission to improve livelihoods and promote environmental sustainability.

To address specific technical needs related to clean cooking technologies, TANJA has consulted ICS manufacturers and, through their introductions, identified authorized repair agents/service centers in Tanzania. Also, TANJA will establish service protocols with these manufacturer-designated agents to handle after-sales support (e.g., repairs, warranty handling, and spare parts), and will include the service network and contact details in user materials.

Where expertise gaps exist, TANJA intends to recruit experienced professionals to oversee key aspects of the project, including a project coordinator with a background in energy efficiency or carbon credit programs. Implementation capacity will be ensured through the appointed manager, trained field teams and, where necessary, contracted specialists for monitoring and data QA/QC.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified.	
Risks to stakeholder participation	No risk identified.	
Working conditions	No risk identified.	
Safety of women and girls	The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	Mitigation measures for gender equality should include promoting equal participation and representation of all genders in project planning, decision-making, and benefits distribution. Additionally, to address gender-specific needs and challenges, TANJA staff will, during regular follow-up/monitoring visits, also promote awareness of participants' safety and well-being.

Safety of minority and marginalized groups, including children	No risk identified.	
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified.	The project replaces traditional three-stone stoves with improved ICS; it does not change fuel use nor introduce any additional pollutant emissions.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ³	Mitigation or preventative measure(s) taken
Discrimination	-Inequitable access to ICS technology based on gender, class, or ethnicity may occur, especially in rural or marginalized communities. -Women, who are often the primary users of cookstoves, might face challenges in accessing or adopting ICS if discriminatory norms persist.	-Ensure equal access to ICS by promoting inclusion and addressing barriers during regular follow-up/monitoring visits, particularly targeting marginalized groups. -Require suppliers to apply non-discrimination policies.
Sexual harassment	-Women engaged in biomass collection or who are involved in ICS distribution and installation may be vulnerable to sexual harassment. -Supply chain or manufacturing processes related to ICS could expose workers, especially women, to harassment if proper policies are not in place.	-Implement strict anti-harassment policies and provide a clear mechanism for reporting and addressing complaints in all areas related to ICS distribution and production. -Require suppliers to maintain a zero-tolerance policy on sexual harassment.

³ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Equal pay for equal work	Women involved in the production, distribution, or marketing of ICS may face wage disparities compared to their male counterparts.	-Enforce equal pay policies across the entire ICS value chain, ensuring that all workers receive fair compensation for their work regardless of gender. -Promote transparency in wage structures and provide regular audits to monitor compliance with equal pay standards. -Require suppliers to apply equal-pay and non-discrimination policies
Gender equity in labor and work	- ICS projects may inadvertently perpetuate traditional gender roles, reinforcing the notion that cooking is a woman's responsibility without addressing broader gender equity issues. -Women might be excluded from leadership roles or decision-making processes in ICS-related projects.	-Promote gender equity by ensuring that women are included in leadership and decision-making roles in ICS projects. -Require suppliers to ensure equal opportunity in recruitment, training, and promotion.
Forced labor	Labor exploitation in the manufacturing or distribution of ICS could result in forced labor, particularly in areas where labor laws are weak.	-Implement strict labor standards and audits throughout the ICS supply chain to ensure that no forced labor is used. -Provide education and resources to local communities to prevent forced labor practices in all stages of ICS production and deployment during follow-up/monitoring visits. -Require suppliers to prohibit forced/bonded labor
Child labor	The production of ICS or related biomass harvesting may involve child labor, particularly in regions with poor labor regulation enforcement.	-Prohibit the use of child labor in all areas of the ICS project, and work with local authorities to promote child labor laws. -Require suppliers to comply with child labor standards

Human trafficking	Vulnerable individuals, particularly women and children, could be at risk of trafficking if labor conditions around ICS production or deployment are not well-regulated.	-Prohibit human trafficking in all areas of the ICS project, and work with local authorities to promote anti-trafficking laws. -Require suppliers to comply with anti-trafficking laws
--------------------------	--	---

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
Risk of complicity in human-rights abuses by staff or suppliers (discrimination, harassment, forced/child labor etc.) during recruitment, distribution, or monitoring.	TANJA shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights. Also TANJA requires suppliers to comply with these standards.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
The project proponent shall recognize and respect the indigenous people's collective rights to own, use, and develop and control the lands, resources, and territories that they have traditionally owned, occupied or otherwise used or acquired, including lands and territories for which they do not yet possess title.	NA

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
The project proponent shall identify all such sites/matters potentially affected by the Project.	Project does not require infringement of any land rights of landowners around the project area.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	- During the Stakeholder meeting (SHM), it was explained that the benefit-sharing scheme would be reviewed and developed in the future. https://tanja.co.tz/improved-cookstove-initiative/index.html https://tanja.co.tz/improved-cookstove-initiative/Project_summary.pdf - After the SHM, each WEO and VEO (among others) was visited, and the explanation was provided again.
Summary of the benefit sharing plan	- During the SHM, it was explained that the benefit-sharing scheme would be reviewed and developed in the future. - After the SHM, each WEO and VEO (among others) was visited, and the explanation was provided again.
Approval and dissemination of benefit sharing plan	Once the benefit-sharing schedule is finalized, it will be announced to all relevant stakeholders by letter and published on the website. We will then await feedback and finalize it after 30 days.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	- Increased demand for biomass could lead to deforestation if ICS requires more wood or if the wood supply is not managed sustainably. - Introduction of ICS may disturb local wildlife habitats if biomass collection areas expand.	- Promote sustainable sourcing of biomass, including afforestation and reforestation programs. - Encourage diversification of energy sources (e.g., biogas or solar alternatives) to reduce pressure on local biomass resources in the future.
Soil degradation and soil erosion	No risk identified	Promote responsible fuel collection: no cutting live trees etc during follow-up/monitoring visits.

Water consumption and stress	No risk identified	-The project is limited to the distribution and household use of ICS and does not increase water consumption or place additional stress on local water resources.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

Species and habitat	The project's activities have no negative effect on local species or their habitats; in fact, by replacing traditional cooking methods it actively supports forest conservation.
Areas needed for habitat connectivity	Since all project actions are confined to household settings and involve no land-use change or tree clearing, areas critical for habitat connectivity remain unaffected. The reduced demand for firewood also indirectly protects forests and helps preserve ecological linkages.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	-The sourcing of biomass for ICS might involve the destruction of habitats that are critical for rare or endangered species, particularly in forested areas. -Increased human activity, such as biomass collection, could disturb these species' natural habitats, reducing their chances for survival.	-Promote sustainable sourcing of biomass from areas that do not overlap with habitats of endangered species. -Work with environmental experts and NGOs to identify and protect critical habitats while implementing the ICS program. -Establish community education programs about the importance of biodiversity and habitat conservation. -Encourage reforestation efforts and ecosystem restoration in regions that are vital for threatened species.

Areas for habitat connectivity	Biomass collection for ICS could lead to fragmentation of habitats, hindering the movement of wildlife and disrupting natural migration routes.	Monitor the impact of biomass collection on habitat connectivity and adjust practices as necessary to minimize disruption.
--------------------------------	---	--

2.4.2 Introduction of Species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A			

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risk identified	

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
-------------------------------------	-----------------------------	-------	---------

Methodology	VM0050	<i>Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0 dated 09 October 2024.</i> https://verra.org/wp-content/uploads/2024/10/VM0050-EE-and-Fuel-Switch-Measures-in-Cookstoves-v1.0.pdf	1.0
calculation	TOOL30	<i>For calculation of fraction of non-renewable biomass, the below tool is used "TOOL30: Calculation of the fraction of non-renewable biomass",</i> https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf	4.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050	Project activities must introduce efficiency improvements in thermal applications of non-renewable biomass in the following premises: households, community-based kitchens, institutions (e.g. schools) or small and medium-sized enterprises (SMEs).	The proposed project entails the distribution of improved cookstoves, enhancing the thermal efficiency of firewood (non-renewable biomass) usage within households.
	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	The Burn stoves to be installed as part of this project are single-pot wood-burning cookstoves with a reported efficiency of 50.8% according to the manufacturer's specifications.

	Both 'Projects' and 'Large Projects' can use the methodology	The proposed project is anticipated to achieve an annual reduction of 3,250 tonnes of CO ₂ . As such, the project meets the criteria for classification as a "Project."
	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	It has been demonstrated that non-renewable biomass has been in use in Tanzania since December 31, 1989.
	For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Not applicable. The ICS is implemented as an energy efficiency measure to replace baseline stoves and decrease the consumption of non-renewable biomass for combustion.
	The project design shall explain the proposed method for distribution of project devices including the method	Each ICS in this project will be uniquely identified through a combination of the customer's name,

	to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo)	geographical location, and a serial number. The serial number will be distinct for each stove, ensuring clear differentiation and preventing any duplication within the project. This guarantees that each stove is accounted for only once. Furthermore, the project has been cross-checked against other CDM project activities operating in the country, using databases from the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to confirm that no ICS is included in any other CDM or voluntary project activities.
	The project design shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The stove manufacturers, wholesale providers, and end users will be required to sign an agreement explicitly stating that the PP or its authorized entity will retain sole ownership of the VCUs generated by the project.

Evidence that the non-renewable biomass has been in use since 1989

Non-renewable biomass has been in use since December 31, 1989 as evidenced by various FAO statistical data. The evidence from FAO’s Global Forest Resources Assessments indicates sustained depletion of Tanzania’s forests since before 1989. FRA 2010 shows a net loss of 8.07 million ha ($\approx 403\,000\text{ ha yr}^{-1}$) between 1990 and 2010, while FRA 2020 reports an annual net forest loss of about $421\,000\text{ ha yr}^{-1}$ ($-0.88\% \text{ yr}^{-1}$) during 2010–2020. These data confirm that the woody biomass traditionally used for cooking has been non-renewable for more than three decades.

In addition—and consistent with Section 1.14—the World Bank Group (2024) reports that more than 89% of households in mainland Tanzania still rely on traditional polluting fuels and technologies for cooking, with 84.8% of rural households chiefly using firewood. The project therefore adopts the 84.8% rural figure as the most appropriate baseline indicator for the areas of implementation. Taken together—long-term net forest loss (FAO FRA 2010/2020) and sustained rural dependence on wood fuels (World Bank 2024)—the evidence demonstrates that non-renewable woody biomass has been in use since before and after 31 December 1989.

Sources: Food and Agriculture Organization of the United Nations (FAO). 2010. Global Forest Resources Assessment 2010 – Country Report: United Republic of Tanzania.

Food and Agriculture Organization of the United Nations (FAO). 2020. Global Forest Resources Assessment 2020: Main Report.

World Bank Group. 2024. “Clean Cooking Background Note – Background Paper for the United Republic of Tanzania Country Climate and Development Report (November 2024).”

3.3 Project Boundary

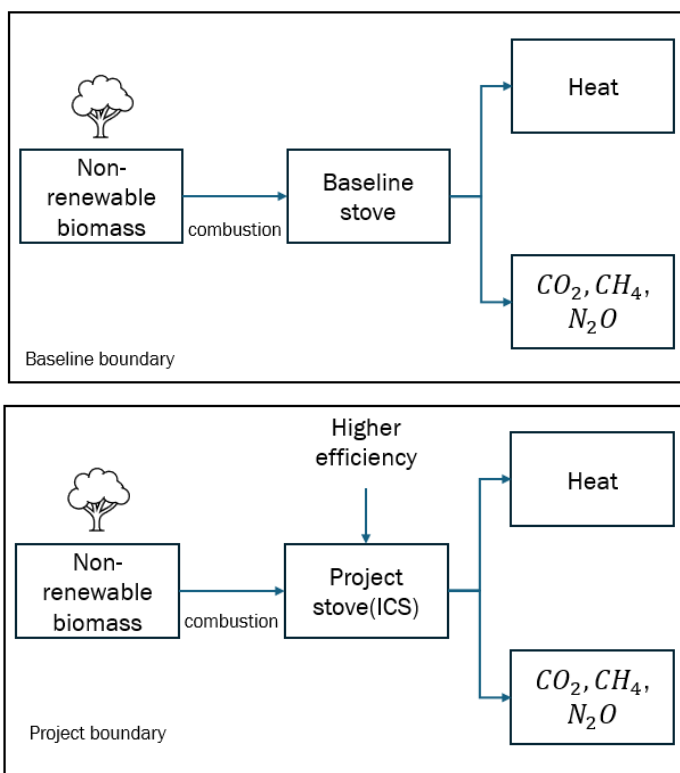
According to VM0050 version 1.0, the project boundary is defined in accordance with the procedure outlined in CDM methodology AMS-II.G version 13.1.

Component : The project boundary encompasses the physical and geographical locations of the efficient devices that utilize biomass.

Project Boundary : For this project, the boundary is defined as the geographical area of Tanzania. For further details, please refer to Section 1.13 (Project Location).

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂	Yes	Major source
		CH ₄	Yes	Major source

Source	Gas	Included?	Justification/Explanation
Project	N ₂ O	Yes	Major source
	Other	No	No other source identified
	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
Source 1	N ₂ O	Yes	Major source
	Other	No	No other source identified
	CO ₂	Yes	Major source
	CH ₄	Yes	Major source



3.4 Baseline Scenario

The baseline scenario assumes the target population would continue using non-renewable wood fuel (firewood) to meet their thermal energy needs in the absence of the project activity. Baseline cookstoves are characterized as three-stone fires or traditional systems without improved combustion air supply mechanisms or flue gas ventilation systems. This project aims to replace these inefficient three-stone fire stoves, which consume large amounts of non-renewable biomass, with high-efficiency stoves. These improved stoves use woody biomass more efficiently, thereby reducing greenhouse gas emissions.

3.5 Additionality

Step 1: Regulatory Surplus

There is no government program or policy in the host country mandating the distribution of domestic fuel-efficient cookstoves for this project. The project is not required by any law, statute, or regulatory framework, including any systematically enforced laws, statutes, or regulatory frameworks applicable to UNFCCC non-Annex I countries.

Participation in this project is entirely voluntary for households. It is confirmed that the proposed project is a voluntary coordinated initiative led by TANJA.

Step 2: Positive List

As outlined in Section 3.2, the project satisfies the applicability conditions of the methodology, which align with the positive list. The project provides ICS to households at a lower price—TZS 10,000 per unit—than both the market price and TANJA's procurement cost and relies on the sale of GHG credits as its revenue source; this end-user payment is below cost and does not generate revenue or profit for TANJA. Additionally, the project is not part of any government programs nor is it funded by multilateral organizations. Conclusion: Since the project meets all the above conditions, it is considered additional.

Step 3: Project Method

This step does not apply to the project activity, as the stoves are distributed to household end-users at TZS 10,000 per unit, lower than both the market price and TANJA's procurement cost. This payment is used to cover distribution, delivery, and operating costs.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☒ Yes ☐ No

3.5.2 Additionality Methods

- The project provides ICS to household end-users at a lower price—TZS 10,000 per unit—than both the market price and TANJA's procurement cost. This end-user payment is allocated to distribution, delivery, and operating costs and it is below cost and does not generate revenue or profit for TANJA. The project therefore relies on the sale of carbon

credits to cover the remaining price gap and the costs of implementation, monitoring, and verification, confirming its additionality.

3.6 Methodology Deviations

- The project does not involve any methodological deviations

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology does not calculate baseline emissions separately; instead, it directly quantifies the net emission reductions achieved by the project. For further details, please refer to Section 4.4.

4.2 Project Emissions

The methodology does not calculate baseline emissions separately; instead, it directly quantifies the net emission reductions achieved by the project. For further details, please refer to Section 4.4.

4.3 Leakage Emissions

Leakage is accounted for using a default value of 0.95, as specified in the methodology.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Net GHG emission reductions are calculated as follows:

$$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y} \quad (1)$$

Where:

ER_y = Emission reductions during year y (t CO₂e)

$LE_{RB,y}$ = Leakage emissions associated with use of renewable biomass during year y (t CO₂e)

Baseline emissions are calculated as follows:

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} + EF_{b,i,nonCO_2}) \quad (2)$$

Where:

BE_y = Baseline emissions during year y (t CO₂e)

$EC_{i,y}$ = Average energy consumption of baseline device type i in year y (TJ)

$N_{j,k,y}$ = Number of commissioned project devices of type j from batch k in year y

$n_{j,k,y}$ = Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)

EF_{b,i,CO_2} = CO₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO₂/TJ)

$f_{NRB,y}$ = Fraction of woody biomass established to be non-renewable, used by the baseline device in year y; not considered for fossil fuels (fraction)

$EF_{b,i,nonCO_2}$ = Non-CO₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO₂e/TJ)

i = Baseline device type and its respective fuel type

j = Project device type and its respective fuel type

k = Batch identifier of the project devices

Project emissions are calculated as follows:

$$PE_y = PE_{energy,y} + PE_{others,y}$$

$$PE_{energy,y} = \sum_j \sum_k BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times \eta_{j,k,y} \times (EF_{p,j,CO_2} \times f_{NRB,y} + EF_{p,j,nonCO_2})$$

Where:

PE_y = Project emissions from energy consumption of project devices in year y

$PE_{energy,y}$ = Project emissions from energy consumption of project devices in year y

$BC_{p,j,k,y}$ = Average quantity of fuel used by project device type j from batch k during year y

$NCV_{p,j}$ = Net calorific value of project fuel used in project device type j

EF_{p,j,CO_2} = CO₂ emission factor for fuel used by project device type j in the project scenario

$f_{NRB,y}$ = Fraction of woody biomass established to be non-renewable, used by the baseline device in year y; not considered for fossil fuels (fraction)

$EF_{p,j,nonCO_2}$ = Non-CO₂ emission factor for fuel used by project device type j

For projects that are not required to assess permanence risk, complete the table below for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-July-2025 to 30-June-2026	531.4	396.8	6.7	127		127
01-July-2026 to 30-June-2027	2,775.2	2,072.3	35.1	667		667
01-July-2027 to 30-June-2028	3,457.7	2,581.9	43.8	832		832
01-July-2028 to 30-June-2029	6,424.8	4,797.4	81.4	1,546		1,546
01-July-2029 to 30-June-2030	11,417.3	8,525.3	144.6	2,747		2,747
01-July-2030 to 30-June-2031	10,846.5	8,099.1	137.4	2,610		2,610
01-July-2031 to 30-June-2032	10,304.2	7,694.1	130.5	2,479		2,479
Total	45,757.1	34,166.8	579.5	11,008		11,008

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	Baseline Energy Consumption (EC _{j,y})
Data unit	TJ
Description	Annual Energy Consumption of Baseline Stove Type
Source of data	Household Survey
Value applied	0.039 TJ
Justification of choice of data or description of measurement methods and procedures applied	Set based on past data
Purpose of data	Baseline Emissions Calculation
Comments	Local field measurement data recommended

Data / Parameter	Number of Households Using Project Stove (N _{j,k,y})
Data unit	HH
Description	Number of Households Using the Project Stoves
Source of data	Distribution Records and Monitoring Data
Value applied	5,000 HH (plan)
Justification of choice of data or description of measurement methods and procedures applied	Based on actual distribution records
Purpose of data	Total Emission Reductions Calculation
Comments	Requires continuous monitoring

Data / Parameter	Emission Factor of Biomass (EF _{b,i,CO2} , EF _{p,j,CO2})
Data unit	tCO ₂ /TJ
Description	Biomass Fuel CO ₂ Conversion Emission Factor
Source of data	IPCC Guidelines
Value applied	112 tCO ₂ /TJ

Justification of choice of data or description of measurement methods and procedures applied	Default data
Purpose of data	CO ₂ Emissions Calculation:
Comments	

Data / Parameter	fNRB,y
Data unit	Fraction
Description	Fraction of Non-Renewable Biomass (fNRB)
Source of data	UNFCCC Report on Updated fNRB Values (2024)
Value applied	0.58
Justification of choice of data or description of measurement methods and procedures applied	Used the latest UNFCCC fNRB data fNRB,y = 0.51 ± 0.07 (Tanzania)
Purpose of data	CO ₂ Emission Reduction Calculation
Comments	Confirmed as the value for Tanzania
Data / Parameter	Emission Factor of Biomass (EF _{b,i,CO2} , EF _{p,j,CO2})
Data unit	tCO ₂ e/t
Description	Biomass Fuel CO ₂ Conversion Emission Factor
Source of data	IPCC Guidelines
Value applied	112 tCO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	Default data
Purpose of data	CO ₂ Emissions Calculation:
Comments	

Data / Parameter	Emission Factor of Biomass (EF _{b,i,nonCO2} , EF _{p,j,nonCO2})
Data unit	tCO ₂ /TJ
Description	Biomass Fuel Non-CO ₂ Conversion Emission Factor
Source of data	IPCC Guidelines
Value applied	9.46 tCO ₂ e/TJ
Justification of choice of data or description of measurement methods and procedures applied	Default data
Purpose of data	CO ₂ Emissions Calculation:
Comments	

Data / Parameter	Biomass Consumption per Household (BC _{p,j,k,y})
Data unit	t/household/year
Description	Biomass Fuel Consumption per Household
Source of data	Kitchen Performance Test (KPT) Measured Data
Value applied	1.7885 t/y (KPT, n=30)
Justification of choice of data or description of measurement methods and procedures applied	Applied locally measured KPT data with a sufficient sample size
Purpose of data	CO ₂ Emissions Calculation
Comments	Further data collection can enhance accuracy

Data / Parameter	Efficiency of Baseline Stove ($\eta_{j,k,y}$)
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch k that remain operating in year y
Source of data	Value will be replaced with the mean of the first biennial survey.

Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	Because the project is still within its first 12 months of operation and no statistically valid household survey has yet been completed, an ex-ante operational proportion of 0.95 is adopted. The parameter will be replaced with the measured mean of the first biennial survey
Purpose of data	CO ₂ Emissions Calculation
Comments	Further data collection can enhance accuracy

5.2 Data and Parameters Monitored

Data / Parameter	N _{j,k,y}
Data unit	Number
Description	Number of Households Using the Project Stoves
Source of data	Installation database and monitoring survey
Description of measurement methods and procedures to be applied	TANJA will keep a comprehensive database of every ICS unit distributed. The rate at which these stoves are actually used will be determined through a representative sample. To set the sample size, the project will apply the latest CDM Standard for sampling and surveys for project activities and programs of activities, targeting 90 % confidence with a 10 % margin of error. Alternatively, TANJA may adopt the simplified method described in Section 8.4, option (b) (page 12) of the methodology, which allows a smaller minimum sample size without requiring adherence to the 90/10 precision threshold.
Frequency of monitoring/recording	At least once every two years
Value applied	5,000
Monitoring equipment	Sampling based monitoring survey
QA/QC procedures to be applied	If the required sampling precision is not attained, the operational fraction of ICS will be derived from conservative lower-bound values rather than repeating the survey.
Purpose of data	Calculation of baseline emissions

Calculation method	The results from sample-based monitoring will be used to calculate $N_{j,k,y}$ as follows $N_{j,k,y} = (n_{j,k,operational} / n_{j,k,total}) * N_{j,k,y,distributed}$ Where: N = number of stoves n = number of samples
Comments	When distributing stoves, teams will check whether the household already owns a project ICS; if so, any second unit provided will be left out of the ER calculations. Alternatively, multiple-stove households can be identified later through sampling, in which case the total ICS count will be reduced by the share of sampled homes using more than one project unit.

Data / Parameter	Life Span
Data unit	Number of years
Description	Project promoters to state the operating lifetime of the project device for projects opting. Please refer to the expected life of the stove stated in Section 1.12 (Manufacturer BURN)
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	All cookstoves will be manufactured to a fixed design specification, guaranteeing uniform dimensions and performance across every unit delivered.
Frequency of monitoring/recording	Once at the time of project stove distributed
Value applied	7-10 years
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Indicate one of the following: • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Calculation method	-
Comments	If additional or new stove models are introduced during the project, their specific thermal-efficiency figures will be applied in the ER calculations. TANJA will either replace a stove at the end of its rated life

	with an equivalent model to continue claiming credits, or else cease crediting once that lifespan has expired.
Data / Parameter	By=1,new,i,survey
Data unit	tonnes/device/year
Description	The annual mass of woody biomass consumed per device type i will be established during the project's first year by means of the implementation of the project through
Source of data	a sample survey
Description of measurement methods and procedures to be applied	Project fuel measurement campaigns/ survey Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, y=1) through fuel measurement campaigns at representative households and/or sample survey.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	For ex-ante calculation purpose, the value is assumed as 1.00 tonnes/device/year
Monitoring equipment	Equipment such as weighing scales and moisture meters will be used for the fuel-measurement campaigns.
QA/QC procedures to be applied	Where the target precision is still not met, results will default to a conservative lower-bound estimate.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Alternatively, questionnaire- or interview-based sample surveys (without fuel-measurement campaigns) may be used to estimate this parameter only if: (i) all baseline stoves have been permanently decommissioned and households rely exclusively on improved cookstoves; and (ii) where several biomass-burning devices are present, the survey clearly distinguishes the amount of woody biomass consumed by the project stove from that used by any other device.

5.3 Monitoring Plan

1. Purpose of Monitoring

In accordance with the VM0050 methodology, this project aims to monitor the fuel-saving effect of introducing high-efficiency cookstoves and accurately calculate the resulting GHG emission reductions. Additionally, a data collection system will be established to enhance the transparency and reliability of the project. TANJA will conduct the following monitoring activities.

2. Monitoring Scope

2.1 Target Region

- **Project area:** Northern Tanzania's Arusha region, covering Oldeani, Daa, Qurus, Ganako, Endamarariek, Endabash
- **Number of project households:** Approximately 5,000
- **Cookstove used:** Ecoa Wood Generation 3

2.2 Monitoring Parameters Required by the Methodology

The following parameters will be monitored according to Section 9.2 of VM0050:

- **N_{j,k,y}:** Number of introduced stoves of type *j* and batch *k*
- **Bold_y:** Annual baseline household fuel consumption (kg)
- **Up_y:** Usage rate of the project stove
- **Pp_y:** Performance retention rate of the stove

3. Monitoring Methods

3.1 Data Collection Frequency

- **Stove introduction status (N_{j,k,y}):** Recorded once a year (frequency to be confirmed)
- **Fuel consumption (Bold_y) and usage rate (Up_y):** Surveyed once a year (after the end of the dry season)
- **Performance retention rate (Pp_y):** Confirmed once a year via sampling

3.2 Data Collection Methods

- **Household Surveys** (questionnaires and observations conducted by field staff)
- **Target:** 10% of the households where stoves have been introduced, randomly selected (minimum 30 households), plus additional backup to reach a total of 40.
- **Data collected:**
 - Weekly average firewood use
 - Stove usage status (frequency of use, satisfaction, alternative cooking methods)
 - Fuel sourcing method (e.g., natural forest, purchase)
- **Method:** In addition to the annual monitoring, TANJA will conduct periodic follow-up visits to verify stove operation, promptly resolve user issues, and improve usage and operational rates.
- **Kitchen Performance Test (KPT)**
 - **Purpose:** Measure the specific fuel-saving effect
 - **Procedure:** Record fuel use for four consecutive days (e.g., Monday to Friday) in selected households
 - **Comparison:** Between baseline (traditional) households and project (improved stove) households
- **Device Distribution Records**
 - **Content:** Track the distribution status of each cookstove
 - **Storage:** Maintain a database with serial numbers, distribution dates, and household information

3.3 Process and Schedule

- **Process and schedule:** Distribution data (Nj,k,y) are recorded continuously at issuance, household surveys and KPT fieldwork are conducted once per year, data entry and cleaning follow promptly after fieldwork, analysis and internal QA are completed thereafter and the Annual Monitoring Report is finalized upon completion of QA. The field team will conduct this per year. The field team conducts these activities every year.

3.4 Instruments, Methods and Calibration

- **Methods and calibration:** Household surveys/KPTs are recorded in digital forms with paper backups where needed. Data are stored in a version-controlled database with weekly backups. Digital scales used in KPTs are calibrated with certified test weights; moisture meters are checked monthly per manufacturer guidance; calibration logs are retained. Aggregation and reporting are performed using a locked calculation workbook with change logs.

3.5 Organization and Responsibilities

- **Organization and responsibilities:** A Monitoring Lead oversees sampling, field logistics, QA/QC and reporting; The Project Manager trains enumerators and perform back-checks; Enumerators collect survey and KPT data and validates datasets and runs ER calculations; the Project Manager approves the report as well.

3.6 Field Protocols for KPT and Surveys

- **Field protocols:** KPT uses digital scales (± 1 g); fuel is air-dried and stored under cover; start, moisture content is recorded for a subsample using a calibrated meter; Each KPT requires four valid consecutive measurement days. If the household is absent on a scheduled day, that day is recorded as missing.

4. Sampling Plan

4.1 Sample Size

- **Following CDM guidelines**, at least 30 households from the target population will be surveyed, plus an additional 10 as backup, for a total of 40.

4.2 Sampling Approach and Precision Targets

- **Sampling approach:** Target precision 90% confidence/10% relative precision for key parameters (Bold,y, Up,y, Pp,y). Sampling is stratified by ward and stove batch; primary sampling units are households selected via random sampling from the distribution database. Sample site coordinates are recorded. Frequency: annual. QA/QC: supervisor spot-observations, back-checks, and re-contact of >10% of sampled households.

5. Data Analysis and Reporting

5.1 Analysis Methods

To calculate GHG emission reductions (ERy), the project will use the formulas in VM0050:

- **ERy:** Annual GHG emission reductions
- **fNRB:** Fraction of non-renewable biomass (e.g., values published by UNFCCC)
- **NCV:** Net calorific value of wood
- **EF:** Emission factor for fuel combustion

5.2 Reporting

An annual monitoring report will include:

- Number of introduced stoves and usage rates
- Reduction in household fuel consumption
- Detailed calculation of GHG emission reductions

6. Data Management and Quality Assurance

- **Data Storage:** Maintain both digital databases and paper copies, ensuring data security and preventing loss
- **Quality Assurance:** Cross-check sampling data, and randomly review survey results to maintain credibility

6.1 Internal Audit and QA/QC

- **Internal audit and QA/QC:** Minimum 10% back-checks of households by the Project Manager; logic checks and outlier screening with call-backs to confirm outliers; double-entry or automated validation for key fields; discrepancy logs with resolutions; annual internal audits of data files and calculations.

6.2 Handling Non-conformances

- **Handling non-conformances:** Deviations (e.g., sample shortfall, equipment failure) are logged, root causes analyzed, and corrective actions implemented. Where re-sampling is impossible, conservative defaults per VM0050 are applied and disclosed in the monitoring report.

6.3 Data Flow and Storage

- **Data flow:** Household/field → Enumerator device → Supervisor review → Project database (weekly backup) → QA/QC checks → ER calculation workbook → Internal audit → Annual Monitoring Report.

6.4 Data Integrity and Protection

- **Data integrity and protection:** Role-based access controls are applied to the database and calculation files; personal identifiers are stored separately from analysis datasets; all changes to the calculation workbook are logged.

7. Risks and Mitigation Measures

- **Risk:** Stove usage rates may be lower than expected
- **Mitigation:** Conduct regular surveys to identify reasons for non-use and promote stove adoption through user education

8. Specific Contributions to the SDGs

SDG 3: Good Health and Well-being

- **Monitoring Guidance:** Health impact assessments
- **Indicator:** Changes in incidence of respiratory illnesses (e.g., coughing, asthma, pneumonia) in surveyed households
- **Method:** Health surveys at baseline and throughout the crediting period; potential data collection from medical or community health centers if available
- **Frequency:** Every two years
- **Measurable Evidence:** Health survey results (evidence of improvement), medical checkup data from local healthcare providers

SDG 5: Gender Equality

- **Monitoring Guidance (1):** Measure time spent collecting fuel
 - **Indicator:** Average time spent collecting firewood per household (hours/week)
 - **Method:** Conduct household surveys on fuel collection time, comparing data before and after ICS introduction
 - **Frequency:** Baseline survey and follow-up every two years
- **Monitoring Guidance (2):** Track changes in how women and children use their time
 - **Indicator:** The percentage of time originally spent on firewood collection now allocated to education, economic activities, household chores, or rest
 - **Method:** Conduct time-use surveys of women and children in target households
 - **Frequency:** Every two years

- **Measurable Evidence:** Time-use survey results (pre- and post-ICS), household interviews

SDG 7: Affordable and Clean Energy

- **Monitoring Guidance:** Measure fuel consumption
- **Indicator:** Annual firewood consumption (kg/year) per household
- **Method:** Conduct Kitchen Performance Tests (KPT) to directly measure cooking fuel usage; record purchase or collection data for each household
- **Frequency:** Baseline survey and annual monitoring
- **Measurable Evidence:** KPT fuel consumption data

SDG 13: Climate Action

- **Monitoring Guidance:** Calculate GHG emission reductions
- **Indicator:** CO₂ emissions reduced (tCO₂eq/year) through decreased fuel use
- **Method:** Use VM0050 methodology to calculate ER based on reduced fuel consumption and the fraction of non-renewable biomass (fNRB)
- **Frequency:** Annual monitoring
- **Measurable Evidence:** GHG reduction calculation reports, recorded fuel savings

Monitoring and Tracking Operations

- **Data Collection and Management:** Assign responsibilities according to each SDG indicator, based on the monitoring plan
- **Use of Digital Tools:** Regularly update and aggregate data in Excel or a similar platform
- **Reporting:** Summarize data annually, and undergo audits as needed to ensure reliability